SYED AFFAN ALI

Mobile & Embedded Systems Engineer | On-Device / Edge AI | Real-Time IoT & Cyber-Physical Systems

affan.syed001@gmail.com • (+92) 307 3086688 • Karachi, Pakistan

linkedin.com/in/syed-affan-dev • *Portfolio: https://syedaffan-saffu.github.io*

PROFILE

Computer Systems Engineering graduate with 1.5+ years of professional experience building mobile and edge applications in Flutter/Dart, with a strong foundation in embedded systems, sensors, IoT, and robotics. Practical experience engineering offline-first applications, background processing with Dart isolates, and hardware-integrated data pipelines that connect sensors, devices, and software. Motivated to pursue graduate research in mobility intelligence and next-generation computing systems — particularly real-time embedded/IoT systems, mobile and edge computing, and efficient on-device AI — bridging hands-on mobile and embedded engineering with systems-level research.

RESEARCH INTERESTS

Real-Time Embedded Systems & Cyber-Physical Systems • Mobile & Edge Computing Systems • On-Device / Edge AI and Lightweight Model Deployment (TinyML) • IoT & Intelligent Sensing Systems • Software for Autonomous & Connected Systems

PROFESSIONAL EXPERIENCE

Mobile Application Developer

Gentec Pvt. Ltd. | Karachi, Pakistan | 11/2024 – Present

- Contribute to production-level in-house mobile application development within cross-functional agile teams, using version control and CI/CD workflows.
- Implement feature modules in Flutter/Dart with an emphasis on clean architecture, scalable state management, and code reusability across platforms.
- Deliver structured tasks within sprint cycles under senior-developer mentorship, participating in daily standups, code reviews, and retrospectives.

Mobile Application Developer – IoT & Connected Applications

Wamtsol Pvt. Ltd. | Karachi, Pakistan | 02/2024 – 10/2024

- Developed and maintained cross-platform Flutter applications with responsive, pixel-accurate interfaces and reusable components for production features.
- Implemented Provider/Bloc state management and structured application architecture to support data-driven features and scalable mobile workflows.
- Integrated device capabilities including GeoLocation, Camera, and media access, building mobile features that interact with device hardware and field data.
- Used Dart isolates for background processing and synchronization, with IsarDB and SharedPreferences for local storage — supporting reliable operation in connected and low-connectivity scenarios.
- Worked on data-oriented mobile workflows relevant to connected-device and IoT use cases, including remote data capture, monitoring, and synchronization.

Flutter Front-End Developer – Connected & IoT Applications

BaijiSoft Technologies | Karachi, Pakistan | 08/2023 – 01/2024

- Translated Figma UI/UX designs into high-fidelity Flutter interfaces for data-driven mobile applications, focusing on reusable components.
- Implemented REST API communication (GET, POST, multipart) to exchange application and device-related data between mobile interfaces and backend services.
- Built data presentation and interaction flows suited to monitoring, status, and control-oriented applications, complementing a background in IoT and embedded systems.
- Ensured consistent Flutter UI behavior across Android/iOS and screen sizes, keeping interfaces practical for field and connected-system use cases.

Mobile Application Developer Intern

Zensoft Pvt. Ltd. | Karachi, Pakistan | 06/2023 – 07/2023

- Built a production Sugar Cane Trolley Registration app in Flutter with native GeoLocation and Image Picker APIs, deployed for real field operations.

- Implemented nested navigation, structured state management, and Google Maps SDK integration for location-aware features.
- Engineered an offline-first background data-upload pipeline using Dart isolates, preserving data integrity without active connectivity.
- Applied unit testing, async programming, and app lifecycle management to deliver a robust, error-resilient application.

EDUCATION

Bachelor of Engineering — Computer Systems
Mehran University of Engineering & Technology (MUET) | Jamshoro, Pakistan | 2019 – 2023

SELECTED PROJECTS

Sugar Cane Trolley Registration App *Flutter · Dart · Dart Isolates · IsarDB · GeoLocation · Image Picker*

- Delivered a production field application for real-time trolley registration with native GeoLocation and Image Picker APIs for geo-tagged photo capture.
- Implemented offline-first architecture with IsarDB local storage and background sync via Dart isolates, ensuring reliable data capture in low-connectivity field conditions.
- Deployed to active field operations, demonstrating end-to-end ownership from requirements gathering through production delivery.

IoT-Based Water Quality Monitoring System *Arduino · Python · Sensors · IoT · Data Visualization*

- Designed and deployed a real-time water quality monitoring system using Arduino microcontrollers with pH, turbidity, and temperature sensors.
- Built a Python data pipeline to collect, normalize, and visualize sensor readings, enabling anomaly detection and threshold-based alerting.
- Implemented wireless data transmission for remote monitoring, demonstrating end-to-end IoT architecture from hardware to dashboard.

Gesture-Controlled Robotic Arm *Arduino · C++ · Servo Motors · Accelerometer · Computer Vision*

- Engineered a gesture-controlled robotic arm using accelerometer-based hand tracking and servo actuation for real-time responsive control.
- Programmed motion-interpolation algorithms in C++ for smooth, precise movement with configurable sensitivity and range-of-motion limits.
- Explored computer-vision-based control as an extension, using OpenCV for hand-landmark detection as an alternative input modality.

AI-Based IoT Energy Monitoring & Anomaly Detection *ESP32 · Python · Machine Learning · IoT · Sensors*

- Built an IoT prototype to collect electrical measurements from an ESP32 and stream readings for remote monitoring and analysis.
- Applied a lightweight anomaly-detection model to identify unusual consumption patterns, presented through a simple monitoring interface.

Smart Greenhouse Monitoring & Prediction System *ESP32 · Python · Sensors · ML · IoT*

- Used temperature, humidity, and light sensors with an ESP32 to collect environmental data for remote monitoring.
- Applied a simple machine-learning model to predict near-term conditions and generate threshold alerts, connecting sensor data with practical decision support.

AI-Powered Document Q&A Agent *Python · LangChain · OpenAI API · FAISS · RAG · Streamlit*

- Built a retrieval-augmented (RAG) agent that ingests documents, creates FAISS vector embeddings, and answers natural-language queries with source citations, deployed via a Streamlit interface.

TECHNICAL SKILLS

Mobile & Edge Flutter · Dart · Provider · Bloc · Dart Isolates · IsarDB · SharedPreferences · REST APIs · JSON · HTTP/Multipart

Embedded & IoT Arduino · ESP32 · C / C++ · Microcontroller Interfacing · Sensor Integration · Wireless Telemetry · Gesture-Controlled Robotics · IoT

AI / ML Python · TensorFlow · PyTorch · Scikit-learn · OpenCV · Object Detection · Real-Time Video Analysis

LLM / Agentic LangChain · RAG Pipelines · OpenAI API · Prompt Engineering

Languages Dart · Python · C · C++ · JavaScript (basic)

Tools & Practices Git · GitHub · Figma · VS Code · Agile/Scrum · Unit Testing · SOLID Principles
Spoken Urdu (Native) · English — IELTS Band 6.5

CERTIFICATIONS & TRAINING

- Mobile App Development with Flutter — MUET | 07/2022 – 12/2022
- Arduino & Robotics — Creativo, Karachi | 04/2021 – 06/2021
- IELTS English Proficiency — Band Score 6.5